Name: ______________________ Neptun code: ______________________

(KTXFI2EBNF) Physics II. Test 1 Solutions
December 1, 2025

1. An electron has a kinetic energy of 5 eV.

   (a) What is its velocity?

   The kinetic energy in SI units is $K = 5 \cdot (1.602176634 \cdot 10^{-19})$ J $= 8.01088317 \cdot 10^{-19}$ J. Using
   $$K = \frac{1}{2}mv^2,$$
   with $m = 9.1 \cdot 10^{-31}$ kg gives
   $$v = \sqrt{\frac{2K}{m}} \approx 1.3269 \cdot 10^6 \text{ m/s}.$$

   (b) What is its momentum?

   $p = mv \approx (9.1 \cdot 10^{-31} \text{ kg}) \cdot (1.3269 \cdot 10^6 \text{ m/s}) \approx 1.2075 \cdot 10^{-24} \text{ kg m/s}.$

   (c) What is the wavelength of the associated de Broglie wave?
   Using $\lambda = h/p$ with $h = 6.62607015 \cdot 10^{-34}$ J s,
   $$\lambda = \frac{h}{p} \approx 5.4876 \cdot 10^{-10} \text{ m} \approx 0.549 \text{ nm}.$$

(10 points)

2. A metal surface is illuminated with light of wavelength $1.1 \cdot 10^{-7}$ m. What is the velocity of the emitted electrons if the photoelectric effect starts at a wavelength of $2.77 \cdot 10^{-7}$ m?

   Use $E_{\text{photon}} = hc/\lambda$ and the kinetic energy of the emitted electron is the excess energy:
   $$K = hc\left(\frac{1}{\lambda_{\text{in}}} - \frac{1}{\lambda_{\text{th}}}\right),$$
   with $h = 6.62607015 \cdot 10^{-34}$ J s and $c = 2.99792458 \cdot 10^8$ m/s. Substituting $\lambda_{\text{in}} = 1.1 \cdot 10^{-7}$ m and $\lambda_{\text{th}} = 2.77 \cdot 10^{-7}$ m gives
   $$K \approx 1.0887 \cdot 10^{-18} \text{ J}.$$
   Then
   $$v = \sqrt{\frac{2K}{m}} \approx 1.5469 \cdot 10^6 \text{ m/s}.$$
   ($m_e = 9.1 \cdot 10^{-31}$ kg)

(10 points)

3. In a Compton scattering experiment, X-rays with a wavelength of 0.1 nm are used.
   ($m_e = 9.1 \cdot 10^{-31}$ kg)

   (a) At what scattering angle does the wavelength of the radiation increase by 1%?
   The Compton wavelength is $\lambda_C = h/(m_e c) \approx 2.4288 \cdot 10^{-12}$ m. An increase of 1% means $\Delta\lambda = 0.01\lambda_0 = 1.0 \cdot 10^{-12}$ m. The Compton formula
   $$\Delta\lambda = \lambda_C(1 - \cos\theta)$$
   gives
   $$1 - \cos\theta = \frac{\Delta\lambda}{\lambda_C} \approx 0.41172,$$
   therefore
   $$\theta = \arccos(1 - 0.41172) \approx 53.97°.$$

(b) At what angle does the wavelength become 0.01% larger?
Now $\Delta\lambda = 0.0001\lambda_0 = 1.0 \cdot 10^{-14}$ m, hence

$$1 - \cos\theta = \frac{\Delta\lambda}{\lambda_C} \approx 0.004117,$$

and

$$\theta = \arccos(1 - 0.004117) \approx 5.20^\circ.$$

(10 points)

4. A light source has a power output of 100 W.
   (a) What is the frequency of the light if the entire energy of the source is emitted as light with wavelength $\lambda = 400$ nm?

$$f = \frac{c}{\lambda} = \frac{2.99792458 \cdot 10^8}{400 \cdot 10^{-9}} \text{ Hz} \approx 7.4948 \cdot 10^{14} \text{ Hz}.$$

   (b) How many photons are emitted by the light source per second?
   Each photon has energy $E = hf$, therefore the photon emission rate is

$$\dot{N} = \frac{P}{hf} = \frac{P\lambda}{hc}.$$

   With $P = 100$ W and $\lambda = 400 \cdot 10^{-9}$ m,

$$\dot{N} \approx 2.0136 \cdot 10^{20} \text{ photons/s}.$$

(10 points)

5. We study a positron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 50 ns. The average lifetime measured in the particle rest frame is 7.5 ns. What is the propagation velocity of the particle beam? Determine the value of $\frac{m}{m_0}$ for the electron. Calculate the rest energy of a positron corresponding to its rest mass $m_p = 9.1 \cdot 10^{-31}$ kg in joules and in electronvolts.

   Time dilation gives $\tau = \gamma\tau_0$, hence

$$\gamma = \frac{\tau}{\tau_0} = \frac{50 \text{ ns}}{7.5 \text{ ns}} = \frac{50}{7.5} = 6.\overline{6}.$$

   Therefore

$$\beta = \frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}} \approx 0.98869,$$

   and

$$v \approx 0.98869\, c \approx 2.9640 \cdot 10^8 \text{ m/s}.$$

   The relativistic mass factor is

$$\frac{m}{m_0} = \gamma \approx 6.6667.$$

   The rest energy of the positron is

$$E_0 = m_p c^2 \approx (9.1 \cdot 10^{-31} \text{ kg}) \cdot (2.99792458 \cdot 10^8 \text{ m/s})^2 \approx 8.1787 \cdot 10^{-14} \text{ J}.$$

   In electronvolts,

$$E_0 \approx \frac{8.1787 \cdot 10^{-14} \text{ J}}{1.602176634 \cdot 10^{-19} \text{ J/eV}} \approx 5.1047 \cdot 10^5 \text{ eV} \approx 510.5 \text{ keV} \approx 0.511 \text{ MeV}.$$

   (Using 1 MeV $\approx 1.6 \cdot 10^{-13}$ J gives $E_0 \approx 0.511$ MeV.) (10 points)

Requirements for obtaining a signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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